American Computer Science League

2021-2022 ● Contest 4: Short Problems ● Junior Division

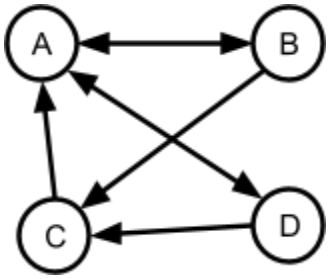
1. Graph Theory

Given vertices {A, B, C, D, E} and edges {AE, AB, BD, BA, CD, DC, BE, EC, CA}, how many paths of length 2 are there?

A. 12
B. 13
C. 14
D. 15
E. 16

2. Graph Theory

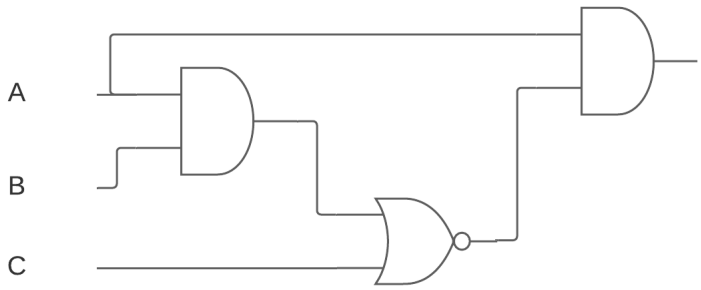
How many cycles are there from A in the directed graph below?



A. 2
B. 3
C. 4
D. 5
E. 6

3. Digital Electronics

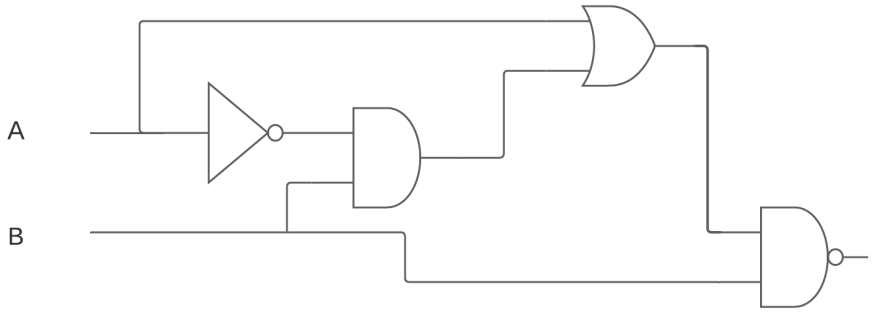
How many ordered triples make the Boolean expression represented by the following digital circuit TRUE?



A. 0
B. 1
C. 4
D. 5
E. 7

4. Digital Electronics

Simplify the Boolean expression represented by the following digital circuit.



A. $A + \overline{B}$
B. $\overline{AB}$
C. $\overline{A + B}$
D. $\overline{A}$
E. $\overline{B}$

5. What Does This Program Do? (Strings)

What is the final length of the string *c* when this program is executed?

```
a = "HURRICANES HENRI AND IDA CAUSED HAVOC FOR MANY"
c = ""
for x = 0 to len(a) - 1
    b = 0
    for y = 0 to len(a) - 1
        if a[x] == a[y] && x != y then
            b = b + 1
        end if
    next y
    if b == 0 then
        c = c + a[x]
    end if
next x
```

A. 1
B. 3
C. 4
D. 8
E. 10

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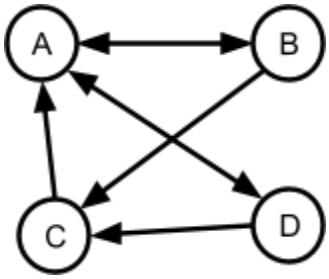
1. 图论

给定顶点 {A, B, C, D, E} 和边 {AE, AB, BD, BA, CD, DC, BE, EC, CA}，请问有多少条长度为 2 的路径？

A. 12
B. 13
C. 14
D. 15
E. 16

2. 图论

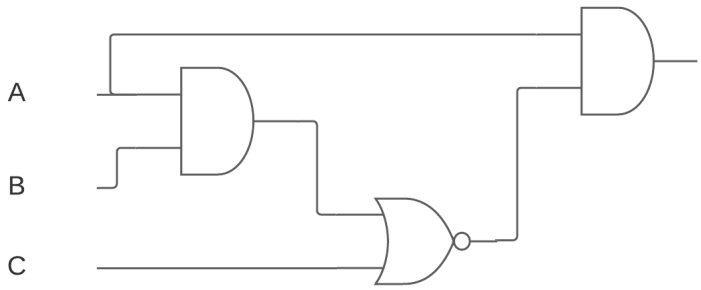
在下述有向图中，请问从 A 出发有多少个环？



A. 2
B. 3
C. 4
D. 5
E. 6

3. 数字电路

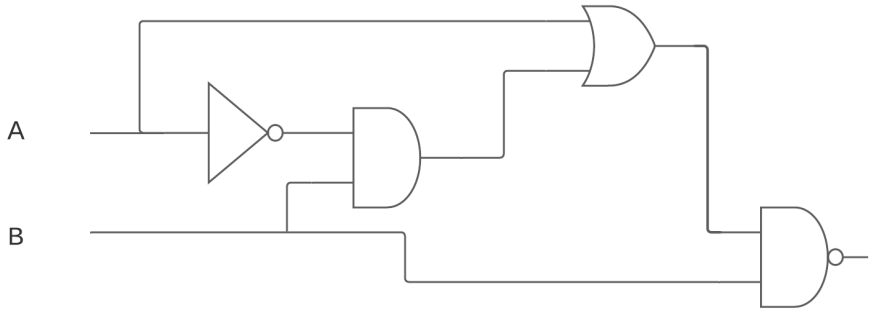
有多少个有序三元组**可以**使得下述数字电路表示的布尔表达式为**真**？



A. 0
B. 1
C. 4
D. 5
E. 7

4. 数字电路

化简下述数字电路表示的布尔表达式。



A. $A + \overline{B}$
B. $\overline{AB}$
C. $\overline{A + B}$
D. $\overline{A}$
E. $\overline{B}$

5. 这个程序的作用是什么？（字符串）

执行以下程序**时**，字符串 *c* 的最终长度是多少？

```
a = "HURRICANES HENRI AND IDA CAUSED HAVOC FOR MANY"
c = ""
for x = 0 to len(a) - 1
    b = 0
    for y = 0 to len(a) - 1
        if a[x] == a[y] && x != y then
            b = b + 1
        end if
    next y
    if b == 0 then
        c = c + a[x]
    end if
next x
```

A. 1
B. 3
C. 4
D. 8
E. 10